



GRAPHIC ERA DEEMED UNIVERSITY

DATA STRUCTURE IN C++ – T360

LibraMind

**An Intelligent Digital Library and Personalized
Book Recommendation System**

PROJECT-BASED LEARNING– PHASE-I

PROJECT AND TEAM INFORMATION

Project Title

LibraMind (An Intelligent Digital Library and Personalized Book Recommendation System)

Student / Team Information

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Course: B.Tech- Computer Science Engineering

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PROPOSAL DESCRIPTION

Motivation

In a college library, finding a useful book is not always easy for every student. Sometimes students know the subject they want to study, but they are not sure which book is good or useful for them. When library have many books, it can take more time to find the right one.

So, we are making LibraMind, which is a digital library system with a basic book recommendation feature. In this system, students can search books, check book availability and also see their borrowing details. Along with this, the system will recommend some books based on things like category, author, subject, ratings and previous reading activity.

The main idea of LibraMind is to make the library more helpful for students. Instead of only searching the book by name, students can also get some related book suggestions according to their interest. In the first phase, we are keeping the recommendation system simple, so we can understand and test how it works properly.

State of the Art / Current solution

Existing library management systems are mostly used for managing books and library records. They can store book information, student details and borrowing or returning records. These systems are useful for normal library work, but they usually do not help much when a student wants to find a new book according to their interest.

Online book platforms have better searching and recommendation features, where users can get suggestions for other books. But our project idea is to bring a simple version of this recommendation feature into a digital library system.

In LibraMind, we are planning to add a recommendation part with the normal library functions. The system can use book category, subject, author, rating, popularity and user's previous reading activity for giving suggestions. In the first version, we will keep it basic and simple. Later, we can improve it by using more advanced Machine Learning and recommendation methods.

Project Goals and Milestones

Our main goal is to make a simple digital library system where students can easily search and find books. User can login ,search books, check book availability and see their borrowing details.

Another goal is to add a recommendation feature in the system so that the system can suggest books according to the student need and interest .We also want to make the system easy to use for both the student and library admin

Milestones:

- **Milestone 1 - Requirements:** First we will decide what features are needed in our library system.
- **Milestone 2 - Database Design:** We will make database for storing users,books,ratings and borrowing details.
- **Milestone 3 - Frontend:** We will create pages for login, search, book details and other user activities.
- **Milestone 4 - Backend:** We will connect frontend with backend using python with flask and handle the main system .
- **Milestone 5 - Recommendation:** We will make a basic recommendation system which will suggest related books to the user.
- **Milestone 6 - Testing:** At last we will connect all the parts and test the system to check if everything works properly.

Project Approach

Front-End

For front-end, we are using HTML, CSS and JavaScript. HTML is used for making basic structure of website. CSS is used for designing the pages and make it look better. JavaScript is used for making website more interactive, like book searching and filtering. It will help user to use the LibraMind system easily.

Back-End

For back-end, we are using Python with Flask. Flask will handle the main working of our system. It will take request from front-end and process it. It will also connect front-end with database and recommendation engine. We are using Python and Flask because it is simple and easy for our project.

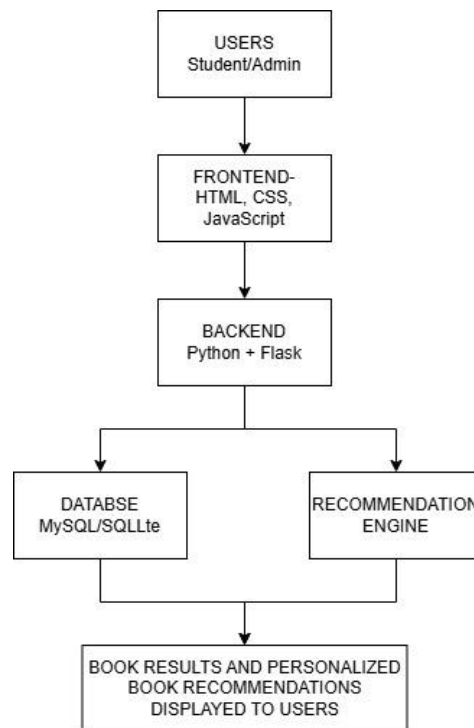
Database

For database, we are planning to use MySQL or SQLite. Database will store important information like books, users, ratings, borrowing history and book availability. This data will be used by back-end whenever it is needed.

Recommendation System

For recommendation system, we will use Book-Crossing Dataset for testing. System will use category, author, subject and ratings of books for giving recommendations. User previous reading activity can also be used for giving better book suggestions. In first version, we will keep recommendation system simple and basic. In future, advanced Machine Learning methods can be added.

System Architecture



The system of LibraMind is divided into different parts. First, user will use the frontend for login, search books, filter books and see book details. The frontend will connect with backend made using Flask. Backend will handle the user requests and connect with database.

Database will store all important information like user details, book details, ratings, borrowing records and book availability. Backend will also connect with recommendation engine. Recommendation engine will check user activity, book category, author, subject and ratings and give suitable book recommendations to the user.

So, the complete flow is:

User → Frontend → Backend → Database + Recommendation Engine → Book Results and Recommendations.

Project Outcome / Deliverables

The main outcome of LibraMind is a working digital library system with recommendation feature.

For Students/Users:

- *User can register and login.*
- *User can search books easily.*
- *User can filter books according to category or other details.*
- *User can see book details and availability.*
- *User can check borrowing and return records.*
- *User can give ratings to books.*
- *User can get personalized book recommendations.*

For Admin/Librarian:

- *Admin can add, update and remove book details.*
- *Admin can manage user information.*
- *Admin can manage borrowing and returning records.*
- *Admin can check book availability and ratings.*

The final deliverable will be a working prototype where normal library functions and book recommendation feature work together. It will also help us understand frontend, backend, database and basic recommendation system in practical way.

Assumptions

We assume that users will provide correct information while registering and will use the system properly. The book details and availability will be updated regularly by the librarian or admin. We also assume that users will give ratings or interact with books, which will help the system understand their interests. Based on this information, LibraMind can suggest books that may be useful or interesting to the user.

Conclusion

LibraMind is a smart library management system that we are planning to develop to make the library experience easier and more useful for students. Along with normal library features like searching books, checking availability, borrowing and returning books, our system will also give book recommendations according to the user's interests and activity.

For developing the project, we are planning to use HTML, CSS and JavaScript for the frontend, Python with Flask for the backend, and MySQL or SQLite for storing the data. The recommendation part will mainly use details like book category, author, subject and ratings to suggest related books.

As this is the first phase of our project, our main focus is to build a simple and working system and understand how the recommendation feature can be added to a library system. In the future, we can improve it by adding Machine Learning, chatbot support, QR/barcode scanning and a mobile application.

References

- *Book Recommendation Dataset (Book-Crossing Dataset) — Kaggle:*
<https://www.kaggle.com/datasets/somesh24/book-recommendation-dataset>